CCIT Report No. 388

An escape time criterion for queueing networks: Asymptotic risk-sensitive control via differential games

Rami Atar, Paul Dupuis and Adam Shwartz

ABSTRACT

We consider the problem of risk-sensitive control of a stochastic network. In controlling such a network, an escape time criterion can be useful if one wishes to regulate the occurrence of large buffers and buffer overflow. In this paper a risk-sensitive escape time criterion is formulated, which in comparison to the ordinary escape time criteria penalizes exits which occur on short time intervals more heavily. The properties of the risk-sensitive problem are studied in the large buffer limit, and related to the value of a deterministic differential game with constrained dynamics. We prove that the game has value, and that the value is the (viscosity) solution of a PDE. For a simple network, the value is computed, demonstrating the applicability of the approach.